12 kW Wind Turbine for Iceland: Best Magnet Shape for N42SH

Status: **study** · 2026-11-05 (Martin: “What would be the most optimal shape for N42SH, 1. in block shape, 2. in fully custom shape, make a comparison with those and the latest 30x15x12”)
Builds on: 19 – Final magnet geometry (Rev B, §16 SH feasibility and the 100 °C design basis), 20 – N42SH 30 × 15 × 6 variant
Analysis: `analysis/generator/optshape.py` → `results/opt_<case>.json`, `opt_summary.json` (26 cases, same 2D nonlinear FEA as docs 19 and 20)

Summary

	30 × 15 × 12 (doc 20)	Best block	Best custom shape
Shape	30 × 15 × 6 blocks, stacked in pairs	Block **33.2 × 14.8 × 11** (one across, five along)	**Arc segment, parallel magnetised**, R 301.5 / 312.5, 6.12° (32.2 mm chord), 11 thick
Line EMF 100 °C, 28 turns (skewed)	365 V	379 V	**387 V**
Turns per coil for 380 V	29	28	28
Copper loss at rated torque vs Rev B	+9 %	±0	**−4 %**
Transient short circuit margin 100 / 120 °C	32 / 13 %	26 / 6 %	30 / 11 %
20 % margin kept up to	**113 °C**	106 °C	110 °C
Magnet mass	8.21 kg	8.21 kg	8.11 kg
Pieces per generator	400	200	160 (4 along)
Cogging not skewed (skewed)	5.4 (0.01) Nm	1.3 (0.00) Nm	**0.5** (0.00) Nm
Drum seat / steel under it	Ø628.0 / 9.0 mm	Ø626.2 / 9.9 mm	Ø625.3 / 10.4 mm, uniform bond
Can be bought	catalogue, stock	made to size, simple	made to drawing, tooling and grinding

Answers:

1. **Best block:** the key for N42SH is a narrow, thick pole, and the 30 × 15 × 12 pair already sits almost at the block optimum for demagnetisation. Only one block keeps the existing 28-turn winding: a custom-size **33 × 15 × 11 mm** block, one across and five along each pole. Compared with 30 × 15 × 12 it brings back 4 % of voltage and 9 % of copper loss, gives up 7 K of short-circuit reserve (106 °C against 113 °C), and as one solid block needs no pair bonding (200 pieces instead of 400).
2. **Best custom shape:** a **parallel-magnetised arc segment**, span 0.68 (6.12°), 11 mm thick, with its back fitted to the round seat. It beats every block:
   - 6 % more voltage than 30 × 15 × 12, so 11 % less copper loss than it and 4 % less than Rev B;
   - 20 % margin kept up to 110 °C;

    - so little cogging (0.5 Nm) that the step skew can be left out;
    - 160 pieces and no stacking.

  The price is a custom ground part: tooling, a minimum order, a longer lead time and a higher piece price.
3. **Loaf shapes (thinner at the edges) are worse for N42SH:** they lose more margin than they gain in smoothness. Radial magnetisation is also worse than parallel.

Recommendation: if a maker outside Asia will quote it, the custom parallel arc (§3.2) is the best N42SH rotor. It is better electrically than Rev B, with only 3 K less short-circuit reserve than 30 × 15 × 12. If only catalogue or plain made-to-size parts are practical, stay with doc 20's 30 × 15 × 12. The 33 × 15 × 11 block is the in-between choice if the 28-turn winding must stay. None of them meets the original 120 °C design case, so every N42SH option keeps the 100 °C basis and the conditions of doc 19 §16.3.

1. What was varied

All cases use N42SH catalogue minimum values (Br 1.29 T, HcJ 1,592 kA/m at 20 °C, −0.12 %/K, −0.60 %/K). Stator, 2 mm gap, 0.5 mm sleeve, magnet gap face at r 301.5 and drum outside r 323 are unchanged. The seat is fitted to each shape, so a thicker magnet takes drum steel away.

Family	What it is	Varied
B	Flat rectangular block, parallel magnetised, one across per pole, five along, on the round seat (bond thicker under the middle)	span 0.63 to 0.80 (width 29.9 to 37.9 mm), thickness 9 to 13 mm
AR	Arc segment, radially magnetised (the doc 18 shape)	span 0.72, 0.80; 9, 11 mm
AP	Arc segment, parallel magnetised	span 0.63 to 0.80; 10 to 12 mm
L	Loaf: arc back, gap face lifted 2 or 4 mm at the edges, parallel magnetised	span 0.80, 0.90; 10 to 12 mm

For each case the FEA gives:

- line EMF at 100 °C with 28 turns, skewed in 4 rows of 0.25°;
- phase THD;
- cogging at 60 °C, not skewed and skewed;
- demagnetisation margin (worst element against HcJ) for rated load and for steady and 2 × transient short circuit, at 100 and 120 °C;
- the temperature that keeps 20 % margin in the transient short circuit, interpolated from those two results.

2. All results

Case	Span	Thick	Mass	EMF 28 t	Cu loss vs Rev B	Phase THD	Cogging (skewed)	SC margin 100 / 120 °C	20 % up to
Blocks									
B63_9	0.63	9	6.05 kg	346 V	+21 %	1.9 %	5.7 (0.0)	28 / 8 %	108 °C
B63_11	0.63	11	7.39 kg	359 V	+12 %	2.0 %	5.5 (0.0)	31 / 13 %	112 °C
B63_13	0.63	13	8.73 kg	367 V	+7 %	2.2 %	5.4 (0.0)	32 / 14 %	114 °C
S1d (doc 20)	0.63	12	8.21 kg	365 V	+9 %	2.2 %	5.4 (0.0)	32 / 13 %	113 °C
B68_11	0.68	11	7.98 kg	374 V	+3 %	4.6 %	1.4 (0.0)	27 / 8 %	107 °C
B70_11	0.70	11	8.21 kg	379 V	±0	5.8 %	1.3 (0.0)	26 / 6 %	106 °C
B70_12	0.70	12	8.96 kg	384 V	−2 %	5.9 %	1.4 (0.0)	27 / 7 %	107 °C
B72_9	0.72	9	6.91 kg	370 V	+6 %	6.5 %	3.2 (0.0)	23 / 2 %	103 °C
B72_11	0.72	11	8.45 kg	384 V	−2 %	6.9 %	3.4 (0.0)	26 / 6 %	106 °C
B72_13	0.72	13	9.98 kg	388 V	−4 %	6.8 %	3.3 (0.0)	27 / 7 %	107 °C
B80_9	0.80	9	7.68 kg	384 V	−2 %	10.7 %	0.6 (0.0)	20 / −1 %	100 °C
B80_11	0.80	11	9.39 kg	398 V	−9 %	10.8 %	0.3 (0.0)	23 / 2 %	103 °C
B80_13	0.80	13	11.09 kg	395 V	−7 %	10.4 %	0.5 (0.0)	25 / 5 %	105 °C
Custom									
AP63_12	0.63	12	8.21 kg	376 V	+2 %	3.0 %	7.1 (0.0)	32 / 13 %	113 °C
AP68_11	0.68	11	8.11 kg	387 V	−4 %	5.9 %	0.5 (0.0)	30 / 11 %	110 °C
AP72_11	0.72	11	8.59 kg	399 V	−9 %	8.3 %	6.8 (0.0)	28 / 9 %	108 °C
AP80_10	0.80	10	8.66 kg	410 V	−14 %	11.9 %	0.4 (0.0)	25 / 5 %	104 °C
AR72_9	0.72	9	7.01 kg	388 V	−4 %	7.6 %	6.4 (0.0)	20 / −2 %	100 °C
AR72_11	0.72	11	8.59 kg	402 V	−10 %	8.0 %	6.5 (0.0)	22 / 1 %	102 °C
AR80_9	0.80	9	7.79 kg	406 V	−12 %	11.6 %	0.5 (0.0)	20 / −2 %	100 °C
AR80_11	0.80	11	9.55 kg	419 V	−18 %	11.5 %	0.7 (0.0)	22 / 1 %	102 °C
L80_10_2	0.80	10	8.10 kg	390 V	−5 %	8.7 %	0.9 (0.0)	22 / 1 %	102 °C
L80_10_4	0.80	10	7.52 kg	369 V	+6 %	5.1 %	0.3 (0.0)	15 / −8 %	96 °C
L80_12_2	0.80	12	9.86 kg	401 V	−10 %	9.0 %	0.8 (0.0)	25 / 5 %	105 °C
L80_12_4	0.80	12	9.29 kg	383 V	−2 %	5.9 %	0.6 (0.0)	19 / −3 %	99 °C
L90_11_4	0.90	11	9.46 kg	394 V	−7 %	8.9 %	0.2 (0.0)	23 / 2 %	103 °C

The copper loss column is the loss at rated torque relative to Rev B. It scales as (380 V / EMF)², because a weaker magnet needs more ampere-turns whatever the turn count.

3. What the numbers say

3.1 Blocks

- **The span sets the trade between voltage and margin.** Going from 0.63 to 0.80 adds about 10 % EMF but costs about 8 points of short-circuit margin (8 K of reserve). In a short circuit the stator's reverse field is strongest at the pole edges, and a narrow pole keeps its edges away from the neighbouring pole and from the stator's d-axis field.
- **Thickness above 11 mm buys little.** At span 0.63, going from 9 to 11 mm adds 4 K; going from 11 to 13 mm adds only 2 K, for 1.3 kg more magnet and 2 mm less drum steel.
- **The block optimum for N42SH is therefore a narrow pole, 11 to 12 mm thick**, which is exactly 30 × 15 × 12 (S1d, 113 °C). Its only weakness is 9 % more copper loss and the 29-turn winding.
- **If the 28-turn winding must stay,** the best block is **span 0.70, 11 mm**: a **33.2 × 14.8 × 11 mm** block, five per pole along the stack.
    - It gives 379 V and Rev B's copper loss with 8.21 kg of magnet, and keeps 20 % margin to 106 °C.
    - It cogs 1.3 Nm unskewed, so the step skew is still advised.
    - It is not a catalogue size, but it is a plain made-to-size block that most makers cut from standard stock without tooling.

3.2 Custom shapes

- **Parallel magnetisation beats radial on both counts.** At span 0.72, 11 mm thick, the parallel arc (AP72_11) gives 399 V with 28 % margin at 100 °C, against 402 V with 22 % for the radial arc (AR72_11). Parallel magnetisation points the material at the pole edges towards the pole centre, away from the reverse field.
- **An arc beats a flat block of the same span and thickness.** At span 0.68, 11 mm thick, the arc (AP68_11) gives 387 V with 30 % margin, against 374 V with 27 % for the flat block (B68_11). The arc fills the corners that a flat block leaves as a thick bond line, its gap face stays at r 301.5 across the whole pole, and the bond is uniform.
- **Loaf shapes are the wrong direction for N42SH.** Thinning the edges lowers cogging and THD, but the thin edges are exactly where the short circuit demagnetises. Lifting the edges by 4 mm costs 7 to 10 K of reserve.
- **AP68_11 is the best balance.**
    - 387 V with 28 turns.
    - Copper loss 4 % below Rev B, 11 % below 30 × 15 × 12.
    - 110 °C reserve, 8.11 kg of magnet.
    - 0.5 Nm cogging unskewed (0.07 % of rated torque). The skew can be dropped, along with the stepped comb bars and the skew template.

  AP63_12 has the same mass and margin as S1d and 3 % more EMF, but it still needs 29 turns and skew.

3.3 Proposed custom part (AP68_11)

Item	Value
Shape	Arc segment, inner R 301.50, outer R 312.50 (11.0 thick), angle 6.12° (chord 32.2 mm at the inner face)
Length	4 pieces along the stack, 18.4 mm each (eddy loss about 6 W)

Item	Value
Grade, magnetisation	N42SH, **parallel** to the segment’s centre line, NiCuNi as doc 19 §8
Count	160 + 8 spares, one part for N and S poles (flipped)
Seat	Ø625.3 (uniform 0.15 bond), drum steel under it 10.4 mm
Comb bars	Plain straight G11 bars, no skew steps (about 15 mm wide at the gap face)
Tolerances (to agree with maker)	Radii ±0.05, thickness ±0.05, angle ±0.05°, length ±0.1

Buying a custom arc means:

- grinding fixtures or press tooling;
- a minimum order, typically a few hundred to a few thousand pieces;
- a longer lead time (my guess: 8 to 12 weeks to samples);
- a piece price above plain blocks.

Neo (Narva) and VAC both make arc segments in SH grades. Ask for it as a second line on the same RFQ.

4. Open items

#	Item
O1	All N42SH options still need the doc 19 §16.3 conditions (bench magnet temperature ≤ 75 °C, temperature watch, no deliberate shorting)
O2	Quote AP68_11 from Neo and VAC (tooling, MOQ, lead time) next to the 30 × 15 × 6 catalogue blocks
O3	If AP68_11 is chosen: geometry module, drawings and a doc 20 style build document; 3D end-effect check
O4	For B70_11 or AP68_11: rerun the full doc 06 load and thermal check (only EMF, cogging and demag were computed here)

Appendix. How to rerun

```
cd analysis/generator
python3 optshape.py B70_11 AP68_11 S1d    # one or more cases, ~6 min each ->
results/opt_<case>.json
python3 optshape.py table                 # -> results/opt_summary.json and a printed table
```